ROI Analyses Electrodes (spheres) - Model 1 (Encode & Recall)

Libraries and Functions

```
# Clear environment
rm(list = ls())

# Get Libraries and Functions
source("~/Dropbox (Personal)/TI_paper/ti-paper/Data/Functions.R")
```

Import data

```
setwd("~/Dropbox (Personal)/TI_paper/ti-paper/Data/MRI/FaceNameTask/ROI/Cortex")
ROIdata <- read.csv("BOLD_Crtx_model1.csv")

# Assign factors
ROIdata$S_ID <- factor(ROIdata$S_ID)
ROIdata$ID <- factor(ROIdata$ID)
ROIdata$ROI <- factor(ROIdata$ROI)
ROIdata$hemisphere <- factor(ROIdata$hemisphere)
ROIdata$StimType <- factor(ROIdata$StimType)
ROIdata$TaskStage <- factor(ROIdata$TaskStage)
```

Sham Condition - Crtx spheres

```
# One sample t-test - Sham Encode, Left hemisphere - Table S10

# BOLD = median % BOLD signal

Sham_enc <- subset(ROIdata, hemisphere=="L" & StimType=="Sham" & TaskStage=="Encode" )
type = "greater"
Sham_ost <- ddply(Sham_enc, .(TaskStage, ROI), t.test.plyr, "BOLD")
Sham_ost <- ost_FDR(Sham_ost)
pander(Sham_ost)
```

Table 1: Table continues below

TaskStage	ROI	statistic	p.value	estimate
Encode	Ant	-0.876333277545527	0.802666637628563	-0.0360975
Encode	Mid	2.05414330514012	0.0289093255675548	0.113165625
Encode	Post	2.56543503179436	0.010764285953679	0.310891875

Table 2: Table continues below

conf.int1	conf.int2	nobs	dof	method	alternative
-0.108308309330272	Inf	16	15	One Sample t-test	greater
0.0165876313118283	Inf	16	15	One Sample t-test	greater
0.0984487200199729	Inf	16	15	One Sample t-test	greater

null.value	sig	FDR
0	NA	0.8027
0	**	0.04336
0	**	0.03229

Stim Conditions - Crtx spheres - Figure 2Q

```
# Does stimulation impact BOLD activity during encode vs recall stages of the task in the regions under

Crtx <- subset(ROIdata, hemisphere=="L")
xtabs(~ TaskStage + hemisphere + StimType + ROI, Crtx)
```

```
## , , StimType = Sham, ROI = Ant
##
##          hemisphere
## TaskStage  L  R
##    Encode 16  0
##    Recall 16  0
##
## , , StimType = TI 1:1, ROI = Ant
##
##          hemisphere
## TaskStage  L  R
##    Encode 16  0
##    Recall 16  0
##
## , , StimType = TI 1:3, ROI = Ant
##
##          hemisphere
## TaskStage  L  R
##    Encode 16  0
##    Recall 16  0
##
## , , StimType = Sham, ROI = Mid
##
##          hemisphere
## TaskStage  L  R
##    Encode 16  0
##    Recall 16  0
##
## , , StimType = TI 1:1, ROI = Mid
```

```
##
##          hemisphere
## TaskStage  L  R
##    Encode 16  0
##    Recall 16  0
##
## , , StimType = TI 1:3, ROI = Mid
##
##          hemisphere
## TaskStage  L  R
##    Encode 16  0
##    Recall 16  0
##
## , , StimType = Sham, ROI = Post
##
##          hemisphere
## TaskStage  L  R
##    Encode 16  0
##    Recall 16  0
##
## , , StimType = TI 1:1, ROI = Post
##
##          hemisphere
## TaskStage  L  R
##    Encode 16  0
##    Recall 16  0
##
## , , StimType = TI 1:3, ROI = Post
##
##          hemisphere
## TaskStage  L  R
##    Encode 16  0
##    Recall 16  0
```

```
BA <- lmer(BOLD ~ StimType*ROI*TaskStage + (1|S_ID), data = Crtx, REML = TRUE)
Anova(BA, test="F")
```

```
## Analysis of Deviance Table (Type II Wald F tests with Kenward-Roger df)
##
## Response: BOLD
##                               F Df Df.res    Pr(>F)
## StimType                 0.8285  2    255  0.437897
## ROI                     21.3390  2    255 2.699e-09 ***
## TaskStage               10.1473  1    255  0.001625 **
## StimType:ROI             0.1870  4    255  0.945032
## StimType:TaskStage       0.3121  2    255  0.732152
## ROI:TaskStage            0.1377  2    255  0.871421
## StimType:ROI:TaskStage  0.1343  4    255  0.969640
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(BA, ddf ="Kenward-Roger")
```

```
## Linear mixed model fit by REML. t-tests use Kenward-Roger's method [
```

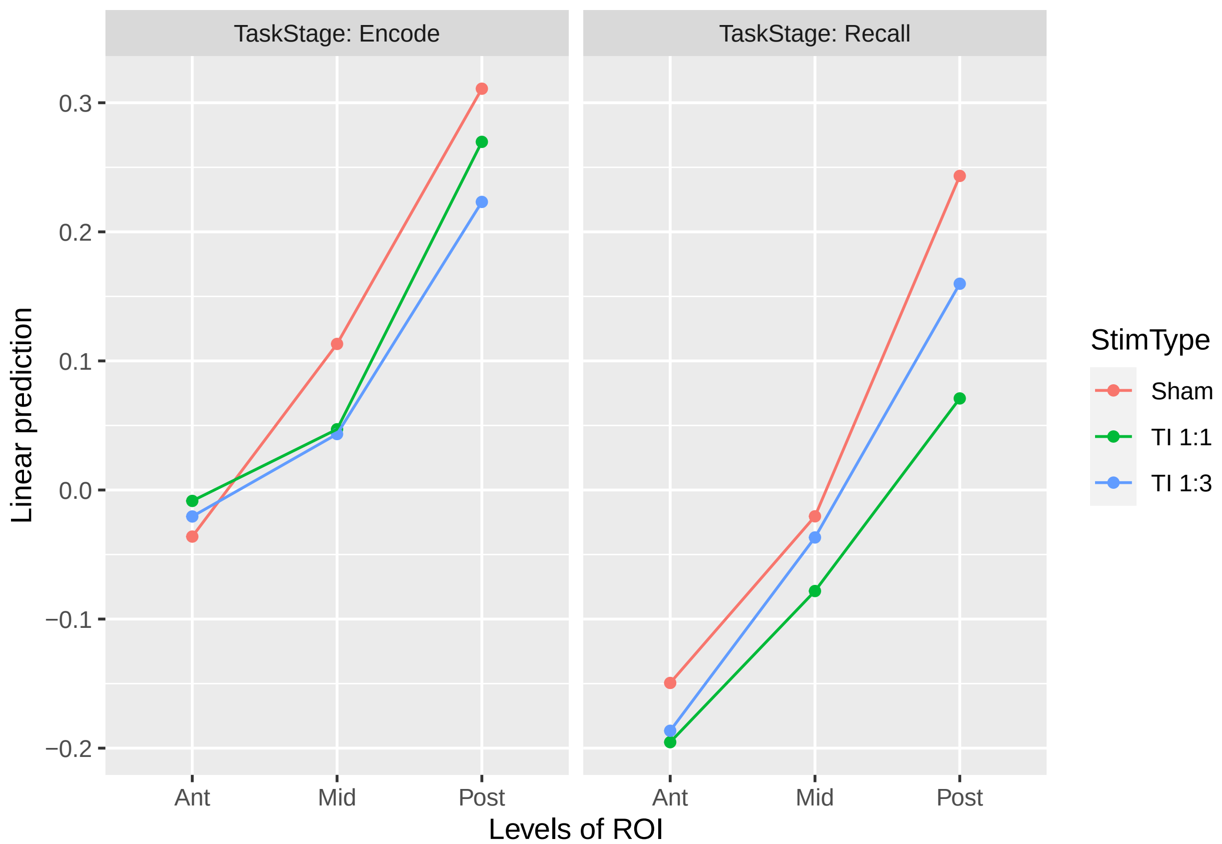
```
## lmerModLmerTest]
## Formula: BOLD ~ StimType * ROI * TaskStage + (1 | S_ID)
##    Data: Crtx
##
## REML criterion at convergence: 244.5
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.3568 -0.6236  0.0087  0.4543  4.3719
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  S_ID     (Intercept) 0.0137   0.1170
##  Residual             0.1129   0.3359
## Number of obs: 288, groups:  S_ID, 16
##
## Fixed effects:
##                                         Estimate Std. Error        df t value
## (Intercept)                             -0.03610    0.08893 225.15522  -0.406
## StimTypeTI 1:1                           0.02763    0.11877 255.00000   0.233
## StimTypeTI 1:3                           0.01557    0.11877 255.00000   0.131
## ROIMid                                   0.14926    0.11877 255.00000   1.257
## ROIPost                                  0.34699    0.11877 255.00000   2.922
## TaskStageRecall                         -0.11339    0.11877 255.00000  -0.955
## StimTypeTI 1:1:ROIMid                   -0.09373    0.16797 255.00000  -0.558
## StimTypeTI 1:3:ROIMid                   -0.08540    0.16797 255.00000  -0.508
## StimTypeTI 1:1:ROIPost                  -0.06882    0.16797 255.00000  -0.410
## StimTypeTI 1:3:ROIPost                  -0.10329    0.16797 255.00000  -0.615
## StimTypeTI 1:1:TaskStageRecall          -0.07353    0.16797 255.00000  -0.438
## StimTypeTI 1:3:TaskStageRecall          -0.05264    0.16797 255.00000  -0.313
## ROIMid:TaskStageRecall                  -0.02017    0.16797 255.00000  -0.120
## ROIPost:TaskStageRecall                  0.04583    0.16797 255.00000   0.273
## StimTypeTI 1:1:ROIMid:TaskStageRecall    0.08174    0.23754 255.00000   0.344
## StimTypeTI 1:3:ROIMid:TaskStageRecall    0.10609    0.23754 255.00000   0.447
## StimTypeTI 1:1:ROIPost:TaskStageRecall  -0.05767    0.23754 255.00000  -0.243
## StimTypeTI 1:3:ROIPost:TaskStageRecall   0.05682    0.23754 255.00000   0.239
##                                        Pr(>|t|)
## (Intercept)                              0.6852
## StimTypeTI 1:1                           0.8162
## StimTypeTI 1:3                           0.8958
## ROIMid                                   0.2100
## ROIPost                                  0.0038 **
## TaskStageRecall                          0.3406
## StimTypeTI 1:1:ROIMid                    0.5773
## StimTypeTI 1:3:ROIMid                    0.6116
## StimTypeTI 1:1:ROIPost                   0.6824
## StimTypeTI 1:3:ROIPost                   0.5391
## StimTypeTI 1:1:TaskStageRecall           0.6619
## StimTypeTI 1:3:TaskStageRecall           0.7543
## ROIMid:TaskStageRecall                   0.9045
## ROIPost:TaskStageRecall                  0.7852
## StimTypeTI 1:1:ROIMid:TaskStageRecall    0.7311
## StimTypeTI 1:3:ROIMid:TaskStageRecall    0.6555
## StimTypeTI 1:1:ROIPost:TaskStageRecall   0.8084
```

```
## StimTypeTI 1:3:ROIPost:TaskStageRecall   0.8111
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
estimated_means <- get_means(BA, "TaskStage*StimType*ROI")
pander(estimated_means)
```

TaskStage	StimType	ROI	Mean	SE	df	CI_lower	CI_higher
Encode	Sham	Ant	-0.0361	0.08893	225.2	-0.2113	0.1392
Recall	Sham	Ant	-0.1495	0.08893	225.2	-0.3247	0.02577
Encode	TI 1:1	Ant	-0.008468	0.08893	225.2	-0.1837	0.1668
Recall	TI 1:1	Ant	-0.1954	0.08893	225.2	-0.3706	-0.02013
Encode	TI 1:3	Ant	-0.02053	0.08893	225.2	-0.1958	0.1547
Recall	TI 1:3	Ant	-0.1865	0.08893	225.2	-0.3618	-0.0113
Encode	Sham	Mid	0.1132	0.08893	225.2	-0.06209	0.2884
Recall	Sham	Mid	-0.02039	0.08893	225.2	-0.1956	0.1549
Encode	TI 1:1	Mid	0.04706	0.08893	225.2	-0.1282	0.2223
Recall	TI 1:1	Mid	-0.07828	0.08893	225.2	-0.2535	0.09697
Encode	TI 1:3	Mid	0.04333	0.08893	225.2	-0.1319	0.2186
Recall	TI 1:3	Mid	-0.03677	0.08893	225.2	-0.212	0.1385
Encode	Sham	Post	0.3109	0.08893	225.2	0.1356	0.4861
Recall	Sham	Post	0.2433	0.08893	225.2	0.06808	0.4186
Encode	TI 1:1	Post	0.2697	0.08893	225.2	0.09445	0.445
Recall	TI 1:1	Post	0.07094	0.08893	225.2	-0.1043	0.2462
Encode	TI 1:3	Post	0.2232	0.08893	225.2	0.04792	0.3984
Recall	TI 1:3	Post	0.1598	0.08893	225.2	-0.01545	0.335

```r
emmip(BA, StimType ~ ROI|TaskStage)
```




```
# Plot
p.BA.Crtx <- ggplot(Crtx, aes(x = ROI, y = BOLD, fill = StimType)) +
       stat_summary(fun = mean, geom = "bar", position = position_dodge(width = 0.9)) +
  stat_summary(geom = "errorbar", fun.data = mean_se, position = position_dodge(0.9), size = 0.6, width
        scale_y_continuous(name = "BOLD (% signal change)",
                           breaks = seq(-1, 1.2, .6),
                           limits=c(-1.2, 1.4)) +
        theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
          axis.text.x = element_text(color = "grey20", size = 14, angle = 45, face = "bold", hjust=1),
        axis.text.y = element_text(color = "grey20", size = 14, angle = 0, face = "plain"),
        axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
        axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
        legend.text = element_text(color="black", size=14, face="plain"),
        legend.background = element_blank(),
        legend.key=element_blank(),
        panel.background = element_blank(),
        legend.title=element_blank()) +
        scale_x_discrete(breaks=c("Ant","Mid","Post"),
        labels=c("Crtx Ant","Crtx Mid","Crtx Post")) +
        theme(panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
     theme(strip.text.x = element_text(size = 14),
          strip.text.y = element_text(size = 20),
          legend.text = element_text(size=14),
          legend.key.size = unit(0.5, 'cm')) + ggtitle("") + xlab("Cortex") + scale_fill_manual(values =
```

```
ggsave(p.BA.Crtx,file="Figure_2Q.tiff", width = 12, height = 9, dpi = 300, units = "cm", device = "tiff
p.BA.Crtx
```